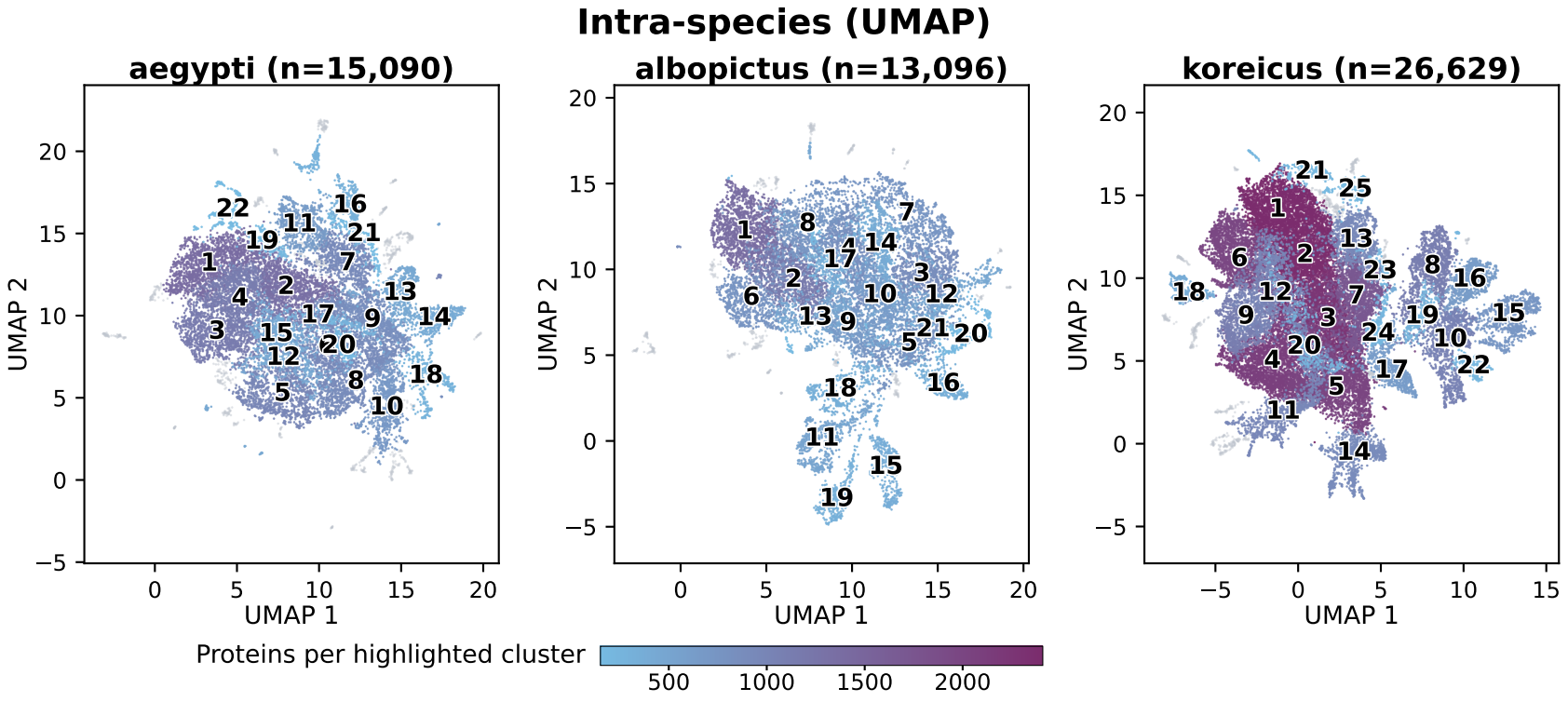
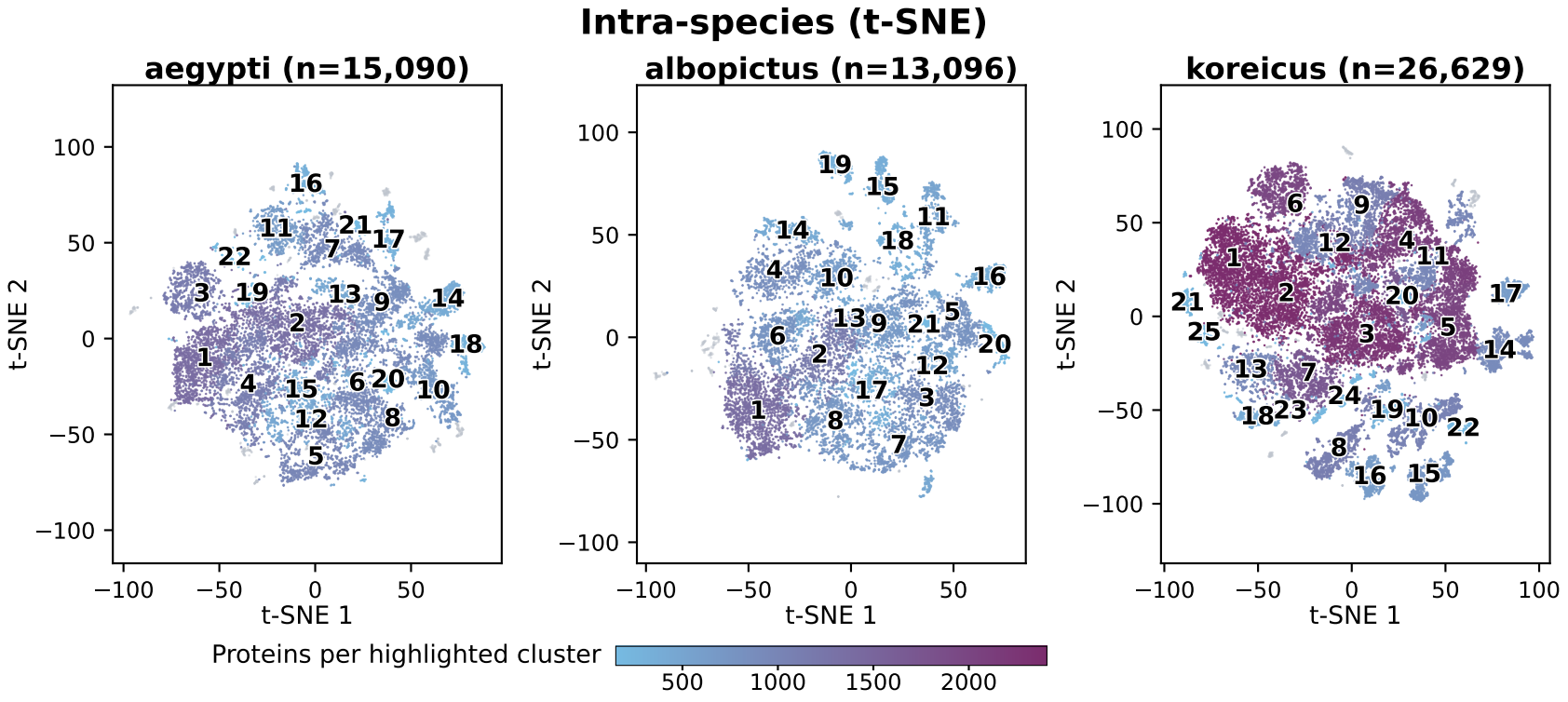
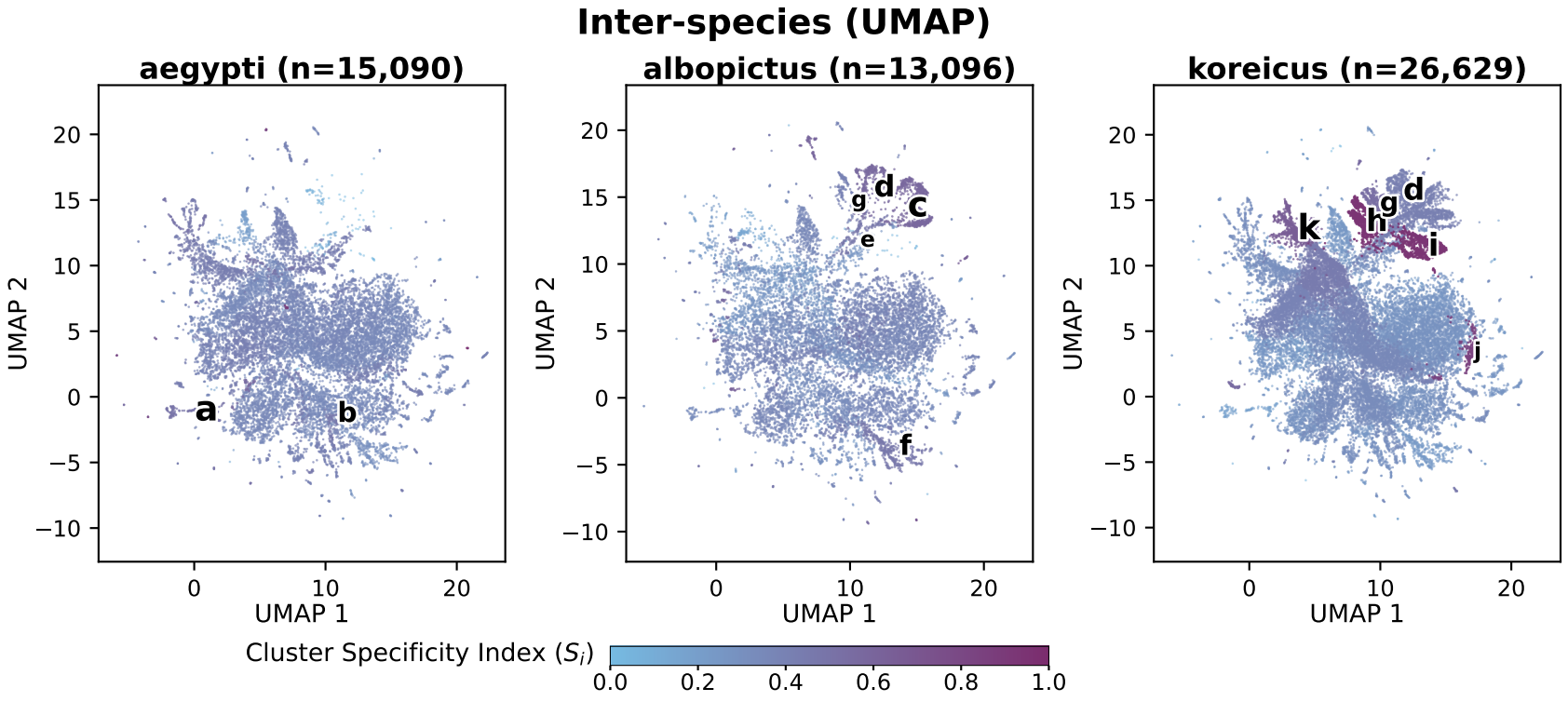
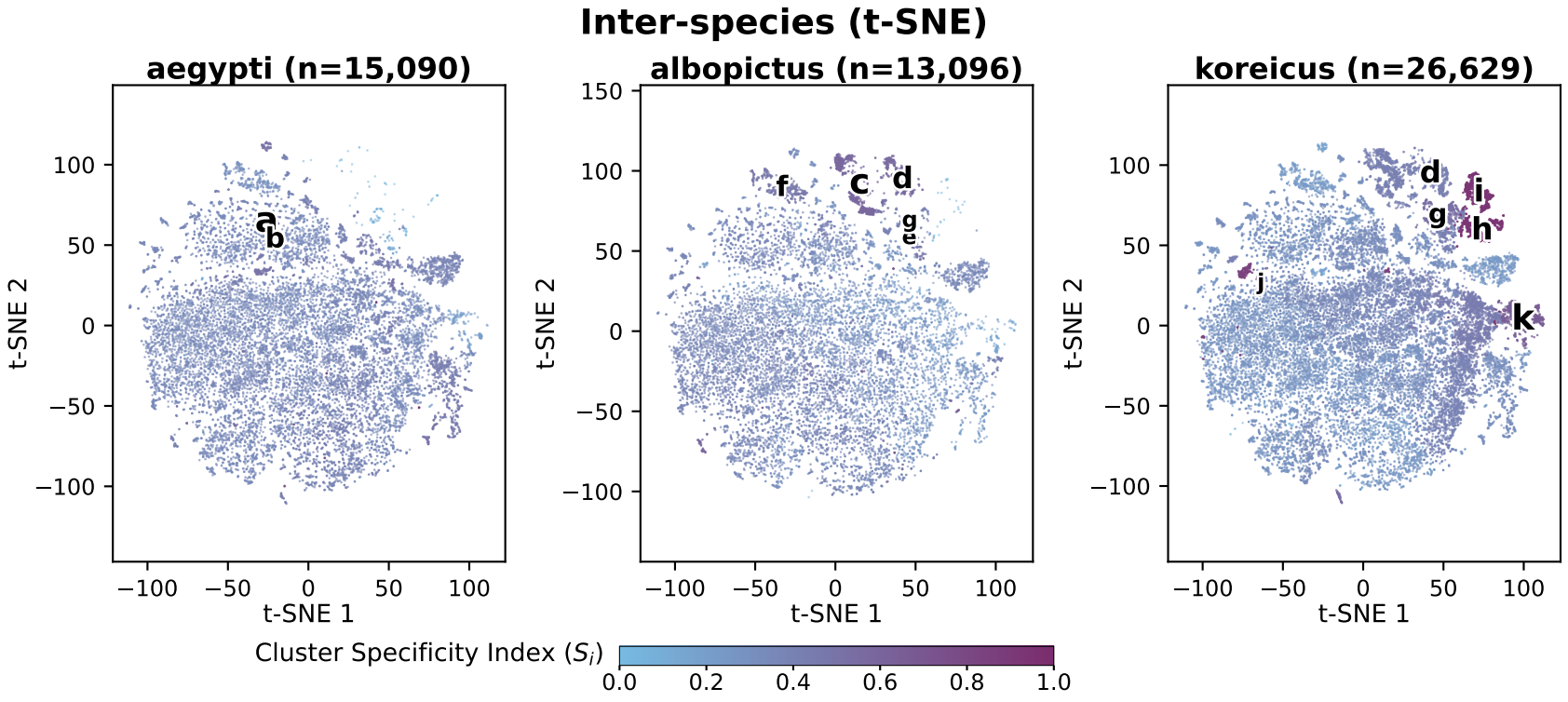
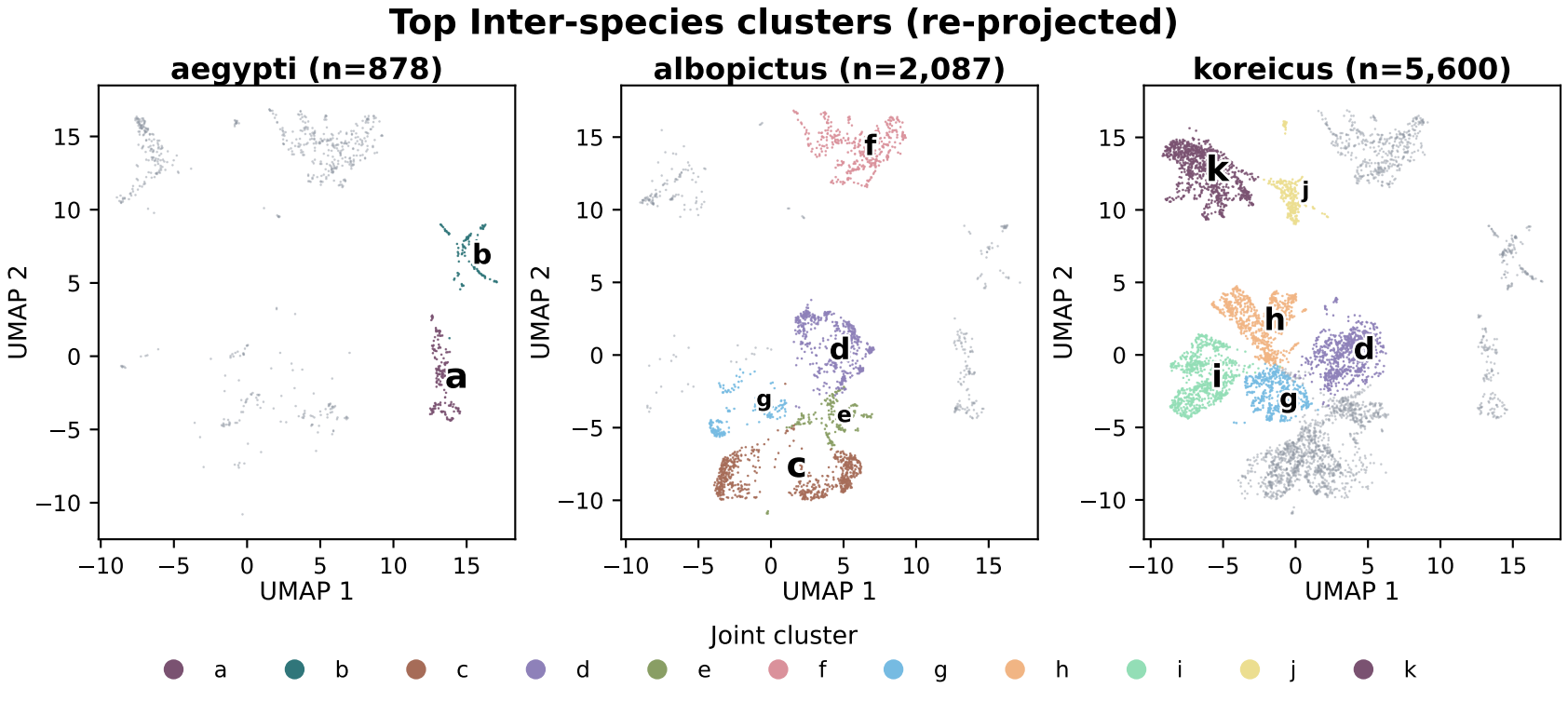
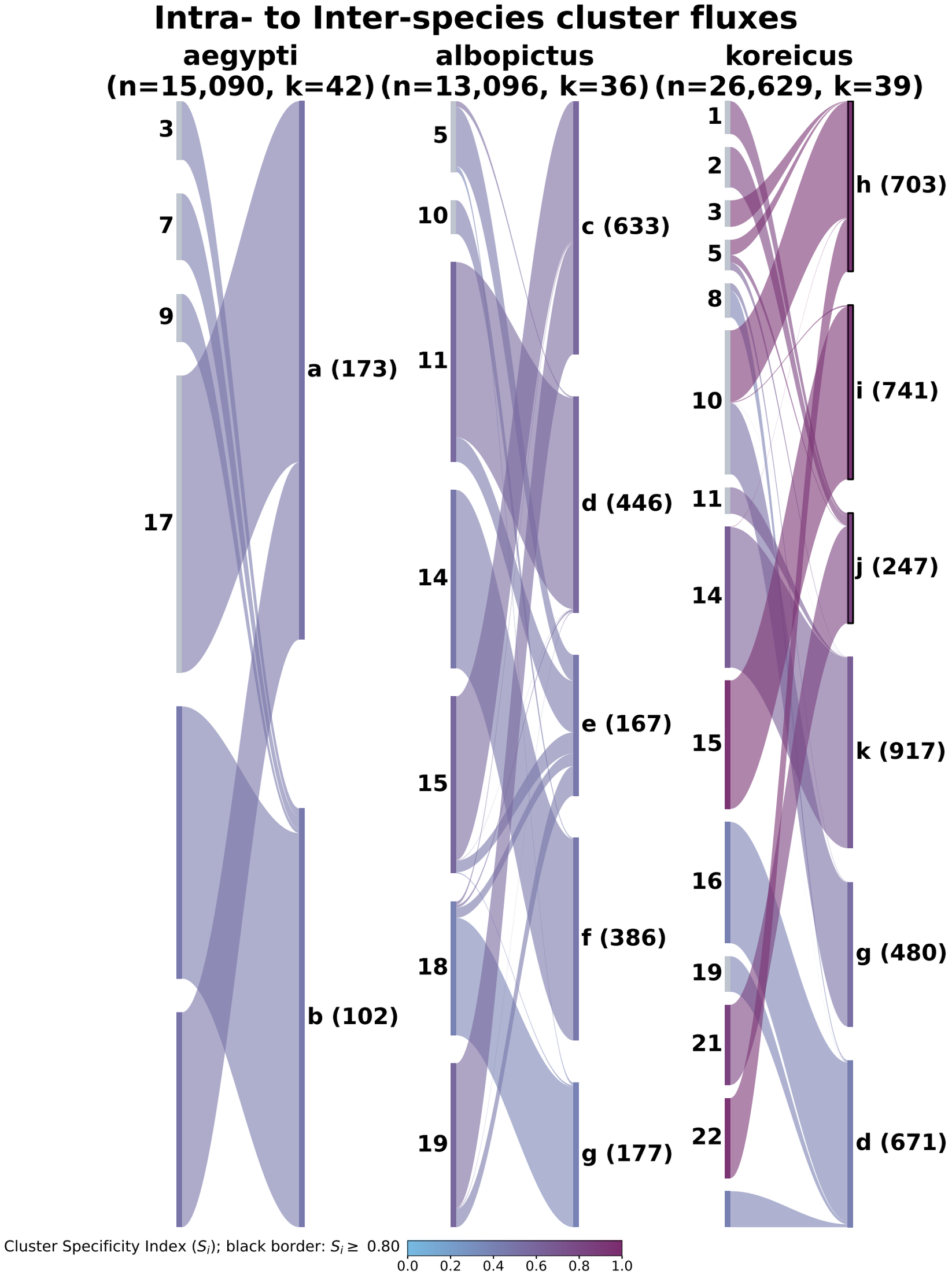

Intra-species (UMAP)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
UMAP 1
UMAP 2
Proteins per highlighted cluster
500
1000
1500
2000

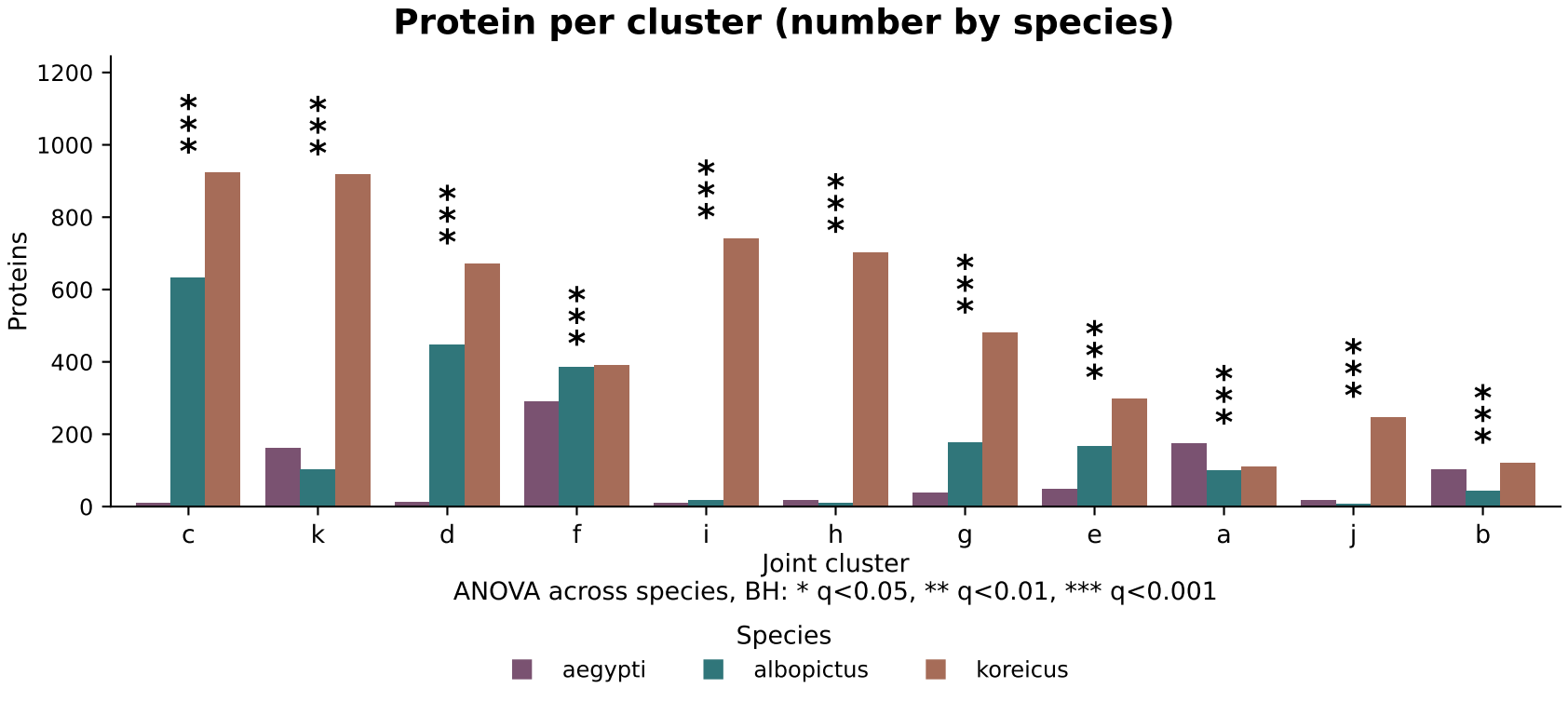

Intra-species (t-SNE)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
t-SNE 1
t-SNE 2
Proteins per highlighted cluster
500
1000
1500
2000


Inter-species (UMAP)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
UMAP 1
UMAP 2
a
b
c
d
e
f
g
h
i
j
k
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Inter-species (t-SNE)
aegypti (n=15,090)
albopictus (n=13,096)
koreicus (n=26,629)
t-SNE 1
t-SNE 2
a
b
c
d
e
f
g
h
i
j
k
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Top Inter-species clusters (re-projected)
aegypti (n=878)
albopictus (n=2,087)
koreicus (n=5,600)
UMAP 1
UMAP 2
Joint cluster
a
b
c
d
e
f
g
h
i
j
k


Intra- to Inter-species cluster fluxes
aegypti
(n=15,090, k=42)
albopictus
(n=13,096, k=36)
koreicus
(n=26,629, k=39)
3
7
9
17
a (173)
b (102)
5
10
11
14
15
18
19
c (633)
d (446)
e (167)
f (386)
g (177)
1
2
3
5
8
10
11
14
15
16
19
21
22
h (703)
i (741)
j (247)
k (917)
g (480)
d (671)
Cluster Specificity Index (S_i); black border: $S_i \geq$ 0.80
0.0
0.2
0.4
0.6
0.8
1.0


Protein per cluster (number by species)
Proteins
1200
1000
800
600
400
200
0
c
k
d
f
i
h
g
e
a
j
b
Joint cluster
ANOVA across species, BH: * q<0.05, ** q<0.01, *** q<0.001
Species
aegypti
albopictus
koreicus